

AirOne Secondary Mission: AirChem-Rad

Vertical profiling of air chemistry, ultraviolet light and ionising radiation in the lowest kilometre of the atmosphere

Overview

AirChem-Rad uses the AirOne CanSat to measure, on one flight, how air chemistry, UV light and background ionising radiation change with altitude. It uses the ascent and the parachute descent. A metal-oxide gas sensor (BME688), a VOC/NOx index sensor (SGP41) and a TVOC/eCO₂ sensor (ENS160) measure air chemistry. A UV-A/UV-B photodiode (VEML6075) and an ambient-light sensor (OPT3001) measure sunlight. A Geiger-Müller counter (DFRobot SEN0463) measures ionising radiation. Every sample is tagged with barometric altitude (BMP581/BME688), GNSS position (MAX-M10S), attitude/spin (BMI270 + MMC5603), a DS3231-backed timestamp and the on-board flight-state machine (mission_state). This gives one time-aligned vertical profile. On the ground we test whether the lowest atmospheric layer (the boundary layer) is chemically different from the air above it, whether UV and radiation follow their expected altitude trends, and how far a low-cost CanSat can detect these effects. The mission reuses the primary-mission sensors and needs no extra hardware or ground-station software changes. All data use the existing JSON convention, and a missing sensor reading is omitted, never sent as zero.

1. Hypotheses (numbered, testable)

Each hypothesis has a measurable quantity, a statistical test and a pass/fail criterion. “Altitude” means altitude_rel (metres above the launch-site baseline that the firmware averages during SELF_TEST).

#	Hypothesis	Measured quantity	Test	Supported if...
H1	VOC concentration falls with altitude, so the BME688 gas resistance rises (cleaner air raises MOX	bme688_gas_resistance, sgp41_voc_raw, ens160_tvoc vs altitude_rel	Spearman ρ , descent leg only	$\rho(R_{\text{gas}}, \text{alt}) > 0$ and $\rho(\text{TVOC}, \text{alt}) < 0$, each with $p < 0.05$

#	Hypothesis	Measured quantity	Test	Supported if...
H2	resistance). If the flight crosses the boundary-layer (BL) top, the chemistry changes as a step , not a smooth gradient, at the same height as the thermodynamic BL top.	Chemistry change-point height z_{chem} vs θ / humidity-gradient height z_{BL}	Piecewise-linear (one breakpoint) fit vs linear fit, compared by ΔBIC	$\Delta\text{BIC} > 6$ favouring the breakpoint model and $ z_{\text{chem}} - z_{\text{BL}} < 50$ m. A null result is valid if z_{BL} is above the apogee.
H3	Radiation count rate depends on altitude. Close to the ground the terrestrial-gamma part falls with height, while the cosmic-ray part slowly rises (roughly exponentially, e-folding over km).	radiation_counts (per-sample increments) binned by altitude	Poisson maximum-likelihood fit of the model in §4.3; likelihood-ratio test vs a constant rate	Constant-rate model rejected at $p < 0.05$. If not, report an upper limit on the gradient.
H4	UV-B rises faster with altitude than UV-A (less ozone/aerosol path), so the UV-B/UV-A ratio increases with height.	veml6075_uvb / veml6075_uva (upward-facing, sunlit samples only)	Weighted linear regression of ratio vs altitude	Slope > 0 at 95 % confidence

#	Hypothesis	Measured quantity	Test	Supported if...
H5	Profiles measured on ascent and descent at the same altitude agree within sensor uncertainty, unless the sensor's response time causes hysteresis .	Same quantity, ascent vs descent, in matched altitude bins	Paired Wilcoxon signed-rank test per quantity	Agreement ($p > 0.05$) for BMP581/BME688 T and P; any significant hysteresis in the gas channels is quantified as a time constant τ
H6	Once the parachute stabilises (≈ 5 s after DESCENT starts), the spin rate settles to a near-constant value. Pendulum/spin in motion modulates the UV and light readings at that frequency.	imu_gyro_z, mag_x/y, opt3001_lux	FFT / Lomb-Scargle of lux vs gyro-derived spin frequency	Lux spectral peak within $\pm 10\%$ of the spin frequency. This confirms the attitude filter used for H4 is needed.

2. Sensors used and what they measure

All fields are sent as field -> {value, unit, sensor_id}. Fields not read successfully in a cycle are **omitted**.

Sensor (sensor_id)	Bus	Telemetry fields (unit)	Role in AirChem-Rad
BME688	I ² C via PCA9548A	bme688_temperature (K),	Temperature and humidity for

Sensor (sensor_id)	Bus	Telemetry fields (unit)	Role in AirChem-Rad
	(0x76)	bme688_pressure (Pa), bme688_humidity (%), bme688_gas_resistance (Ohm)	potential temperature θ and specific humidity q (BL detection). MOX gas resistance is a broad VOC/reducing- gas proxy (H1, H2).
BMP581	I ² C via mux	bmp581_temperature (K), bmp581_pressure (Pa)	Main high- resolution pressure for altitude. Second temperature to cross-check self- heating.
SGP41	I ² C via mux	sgp41_voc_raw, sgp41_nox_raw (ticks); sgp41_voc, sgp41_nox (Sensirion Gas Index, 0–500)	VOC and NOx (oxidising gases). Raw ticks are used for profiles, because the Gas Index adapts to its own baseline (§4.1). The index is sent only once the algorithm has a valid value.
ENS160	I ² C via mux	ens160_tvoc (ppb), ens160_eco2 (ppm), ens160_aqi (1–5), ens160_status (flag)	Second, independent MOX-array estimate of TVOC/eCO ₂ . The status flag marks warm-up/initial- start-up samples, which are excluded.
VEML6075	I ² C via mux	veml6075_uva, veml6075_uv (counts)	UV-A (≈ 365 nm) and UV-B (≈ 330 nm) irradiance proxies (H4).

Sensor (sensor_id)	Bus	Telemetry fields (unit)	Role in AirChem-Rad
OPT3001	I ² C via mux	opt3001_lux (lux)	Visible light. Used to detect sun-facing vs shadowed samples and the spin modulation (H6).
SEN0463 Geiger counter	GPIO interrupt	radiation_counts (cumulative counts), radiation_cpm (CPM, 60 s rolling window)	Ionising radiation (β/γ) (H3). Fits use cumulative counts; CPM is for display only (it smooths over 60 s).
BMI270	SPI (CS = 15)	imu_accel_x/y/z (m/s ²), imu_gyro_x/y/z (deg/s)	Launch detection, spin rate, attitude filter for UV/light samples (H6).
MMC5603	I ² C via mux	mag_x/y/z (μ T)	Heading/spin phase, independent of gyro drift.
MAX-M10S GNSS	UART2	gnss_lat, gnss_lon (deg), gnss_altitude (m)	Horizontal drift (wind profile) and an independent altitude check.
DS3231 RTC	I ² C via mux	Frame-header timestamp (μ s since Unix epoch: GNSS $\rightarrow$ DS3231 $\rightarrow$ 0)	Absolute time when GNSS has no fix. Needed to compare with weather and solar-elevation data.
INA219	I ² C via mux	battery_voltage (V), battery_current_m a (mA)	Housekeeping. Large current steps (heater cycles of the MOX sensors) are flagged so

Sensor (sensor_id)	Bus	Telemetry fields (unit)	Role in AirChem-Rad
Flight-state machine (FSM)	—	altitude_rel (m), vertical_speed (m/s), mission_state (string), mission_state_code (0-6), sensor_health_mask (bitmask)	they are not mistaken for real chemistry changes. Splits the flight into BOOT, SELF_TEST, PRELAUNCH, ASCENT, APOGEE, DESCENT and LANDED for analysis (codes 0-6 in that order).
Note on mission_state. The unchanged ground parser accepts only numeric values, so it marks the string field mission_state invalid but still accepts the frame. Ground analysis therefore uses mission_state_code (0 = BOOT ... 6 = LANDED). The string is kept for people reading the raw logs.			

3. Expected altitude profiles

The expected flight envelope is a rocket- or drone-launched CanSat reaching a few hundred metres to ≈ 1 km, with a parachute descent of tens of seconds to a few minutes. The values below are expected **trends**, not calibrated predictions.

Quantity	Expected profile with altitude	Physical reason	Main confounders
Pressure (*_pressure)	Falls ≈ 12 Pa per metre near the surface (hypso-metric)	Hydrostatic balance	None significant. This is the altitude reference.
Temperature	Falls ≈ 6.5 -9.8 K/km. It may rise with height through a surface or BL-top inversion.	Adiabatic cooling / inversions	Solar heating of the can, sensor self-heating. Cross-check BMP581 against BME688.
Potential	About constant	Mixing within	—

Quantity	Expected profile with altitude	Physical reason	Main confounders
temperature θ	inside a well-mixed daytime BL, then a sharp rise at the BL top	the BL	
Humidity / specific humidity q	About constant in the BL, then drops above the BL top	Surface moisture source, trapped below the inversion	RH sensor lag on fast ascent
bme688_gas_resistance	Increases with altitude, with a step increase at the BL top	Fewer reducing VOCs aloft	Strong dependence on humidity and temperature. Corrected in §4.1.
sgp41_voc_raw / ens160_tvoc	Decrease with altitude, with a step decrease at the BL top	Surface VOC sources (traffic, vegetation) mixed only within the BL	Sensor response time (seconds to tens of seconds) causes ascent/descent hysteresis (H5)
sgp41_nox_raw	Small decrease, possibly within noise at low NO ₂	Surface NO _x sources	Low sensitivity at background levels
ens160_eco2	Weak decrease. This is a TVOC-derived equivalent , not real CO ₂ .	Calculated from MOX response	Should not be read as CO ₂
veml6075_uvb, veml6075_uva	Both increase slightly (a few % per 100s of m). UV-B/UV-A increases.	Shorter path through ozone/aerosol, less haze below the BL top	Attitude/spin (sensor not sun-facing), clouds. Strong filtering needed (§4.4).
opt3001_lux	Strongly modulated by spin. The envelope rises slightly with altitude above haze.	Aerosol extinction	Spin and pendulum motion

Quantity	Expected profile with altitude	Physical reason	Main confounders
Radiation count rate	Near the ground: falls over the first ~100–300 m as the terrestrial γ -ray contribution (^{40}K , U/Th series) is absorbed by air. Higher up: a slow rise from the cosmic-ray secondary component. Over < 1 km the net change is small (tens of %).	Air absorbs ground radiation; the cosmic-ray flux grows with height up to the Pfotzer maximum at $\approx$ 15–20 km, which is far above CanSat altitude	Low count statistics. At tens of CPM, a 2–3 minute flight gives only tens of counts in total (§4.3).
Spin rate (imu_gyro_z)	High and irregular at ejection, then settles to a near-constant rate within seconds	Parachute aerodynamics	—

4. Data products and analysis plan

4.1 Pre-processing (ground side)

1. **Load** frames logged by the unchanged ground station (CSV/JSON export). Check the frame CRC (already done by the ground protocol layer) and drop duplicate sequence numbers. The firmware's FRAM keeps seq increasing across a brownout reset, so gaps in seq show lost packets, not reboots.
2. **Segment** by mission_state_code. Profiles use ASCENT (3) and DESCENT (5). PRELAUNCH (2) gives the ground reference and sensor warm-up check. LANDED (6) gives a second ground reference.
3. **Altitude.** Use altitude_rel from the on-board filter as the main altitude. Recompute it independently from bmp581_pressure with the hypsometric equation, using the measured surface temperature. Use gnss_altitude only as a sanity check (its vertical error is ~several m).
4. **Derived variables:**
 - Potential temperature $\theta = T \cdot (100000 / p)^{0.2857}$

- Specific humidity q from RH, T and p (Magnus formula)
 - Humidity-corrected gas resistance: fit $\ln R_{\text{gas}} = a + b \cdot q + c \cdot T$ on the PRELAUNCH segment, then use the residual in flight. This takes out the known humidity/temperature sensitivity of MOX sensors.
5. **Quality flags:**
- Remove ENS160 samples whose ens160_status is not “normal operation”.
 - Remove SGP41 *index* values during the algorithm’s start-up blackout. Raw ticks are always kept.
 - Use sensor_health_mask to remove samples from sensors that dropped off the bus.
 - Flag samples with a battery-current step > 50 mA.

4.2 Products

Product	Description	Format
P1 Time series	Every field vs time, with flight phases shaded	PNG + CSV
P2 Binned vertical profiles	Each quantity averaged in 10 m altitude bins (20 m for UV, 50 m for radiation), ascent and descent separately, with median $\pm$ IQR and N per bin	PNG + CSV
P3 Correlation matrix	Spearman ρ between every science quantity and altitude, with p-values (Holm-Bonferroni corrected for 6 hypotheses)	Table
P4 Boundary-layer report	z_{BL} from θ and q gradients, z_{chem} from change-point fits, with confidence intervals	Table + annotated profile plot
P5 Radiation fit	Fitted model parameters, likelihood-ratio statistic, counts per bin with Poisson error bars	Table + plot
P6	Ascent-minus-descent	Table

Product	Description	Format
Hysteresis/response-time	differences and fitted first-order τ for each gas sensor	
P7 Flight track	GNSS ground track and drift-derived wind estimate vs altitude	Map/plot

4.3 Radiation (CPM-vs-altitude) fit

- **Counts, not CPM.** The firmware's radiation_cpm is a 60 s rolling window, which smooths over ~ 60 s and correlates neighbouring samples. Fit instead the per-sample **increments** Δ_n_i of radiation_counts over intervals Δ_t_i . These are independent Poisson variables.
- **Model:** $\lambda(z) = A \cdot e^{(-z/L_\gamma)} + B \cdot e^{(z/H_c)} + C$.
 - The first term is terrestrial γ ($L_\gamma \approx 100$ -200 m in air).
 - The second term is cosmic ($H_c \approx 1$ -2 km).
 - C is the tube's intrinsic background.
 - Over a short flight the model is too flexible, so L_γ and H_c are **fixed** at literature values and only A and B are fitted. C is constrained by a long ground measurement before launch.
- **Likelihood:** $\ln \mathcal{L} = \sum_i [\Delta_n_i \ln(\lambda(z_i)\Delta_t_i) - \lambda(z_i)\Delta_t_i]$. Maximise it, and compare with the constant-rate model using the likelihood-ratio statistic (χ^2 , 1 d.o.f.).
- **Honest statistics.** At ~ 20 -30 CPM background and a flight of ~ 2 -3 minutes, the total is roughly 40-90 counts. That is enough to detect a change of more than ~ 30 -50 % between the bottom and top halves of the profile, but not small gradients. The report therefore gives either a detection or a **90 % upper limit** on the gradient. To improve the statistics:
 1. Combine ascent and descent.
 2. Run a long pre-launch ground baseline (≥ 30 min, $\sim 10^3$ counts).
 3. If allowed, add a tethered-drone or tall-building calibration profile using the same firmware.

4.4 UV analysis

- Use only samples where the payload is roughly sun-facing: `opt3001_lux` is in the top 20 % of its rolling 2 s window, **and** the tilt from `imu_accel_*` is $< 30^\circ$.
- Compute the UV-B/UV-A ratio from these samples only.
- Do the regression against altitude, weighted by $1/\text{variance}$ per bin.
- Compare the result with the solar elevation at the frame timestamp, so the ratio is not confused with a change in sun angle (which is negligible over a few minutes).

4.5 Uncertainty budget

Combine the following in quadrature for every binned profile:

- Sensor datasheet accuracy
- Bin standard error
- Altitude error (BMP581 $\approx \pm 0.5$ m relative over the flight, plus temperature-assumption error in the hypsometric equation)
- Response-time smearing ($\tau \times \text{vertical speed}$)

The response-time smearing usually dominates for the gas sensors. Ascent at tens of m/s gives far worse vertical resolution than descent at $\sim 5\text{--}10$ m/s, so **descent is the primary science leg** for H1, H2 and H4.

5. Ground-side analysis methods

5.1 Correlation with altitude

- **Pearson r** (linear) and **Spearman ρ** (monotonic, robust to sensor non-linearity), computed on the descent leg for each science quantity vs `altitude_rel`.
- Autocorrelation in a time series inflates significance. The **effective sample size** is therefore estimated as $N_{\text{eff}} = N(1 - r_1)/(1 + r_1)$, where r_1 is the lag-1 autocorrelation, and N_{eff} is used for p-values.
- Bootstrap (block bootstrap, 5 s blocks) 95 % confidence intervals are reported for ρ .

5.2 Boundary-layer detection

Use three independent estimators and report all three:

1. **θ -gradient method:** z_{BL} is the lowest height where $d\theta/d_z > 3.5$ K/km after smoothing with a 30 m running median. This is the parcel/gradient method used for radiosondes.
2. **Humidity method:** z_{BL} is the height of the most negative d_q/d_z .
3. **Chemistry change-point:** fit a two-segment piecewise-linear model to humidity-corrected $\ln R_{gas}$, $sgp41_{voc_raw}$ and $ens160_{tvoc}$, searching for the breakpoint z_{chem} on a 5 m grid. Compare with a single linear fit using BIC ($\Delta BIC > 6$ means strong evidence for a step).

H2 is tested by checking whether z_{chem} agrees with methods 1 and 2. If no θ inversion appears below apogee, the BL top is above the flight ceiling. That is recorded as an informative null result, together with the context: time of day, cloud base, and the nearest radiosonde or model BL height if available.

5.3 CPM-vs-altitude fit

Follow §4.3: a Poisson maximum-likelihood fit on count increments, a likelihood-ratio test against a constant rate, and a profile-likelihood 68 %/90 % interval on B/A . As a cross-check, plot the bin-averaged CPM (50 m bins) with $\sqrt{N}$ error bars, with the fitted curve overlaid.

5.4 Ascent/descent hysteresis

- For each gas channel, model the sensor as a first-order system $y_{meas}(t) = y_{true}(z(t)) * (1/\tau)e^{(-t/\tau)}$.
- Fit τ so that the ascent and descent profiles collapse onto one curve.
- Report τ , and deconvolve the descent profile with it before running the H1/H2 tests.

5.5 Implementation

- Analysis scripts in Python (NumPy, SciPy, pandas, matplotlib) read the ground station's existing exports. The ground software already parses `bme688_gas_resistance`, `battery_current_ma` and `mission_state_code` as ordinary numeric fields, so the **ground station software needs no changes**.
 - The same scripts run on the PRELAUNCH segment and on a bench test with a hand-moved "flight", so the pipeline is validated before launch day.
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6. Relevance to the UK CanSat Competition

- **Fits the competition structure.** The UK CanSat Competition requires every team to do the common **primary mission**: measure air temperature and air pressure and transmit them to the ground station during descent. Each team also designs its own **secondary mission**. AirChem-Rad uses the primary-mission measurements (BMP581/BME688 temperature and pressure) as its altitude backbone and adds a scientific question on top. Teams should check the current year's official guidelines for exact requirements (mass, size, telemetry rate, descent rate, recovery).
 - **Scientific merit judged by the panel.** The mission has clear hypotheses with stated pass/fail criteria, an uncertainty budget, and a planned null-result interpretation. That shows the scientific method, rather than “we logged lots of sensors”.
 - **Real-world link.** Boundary-layer air quality affects urban pollution forecasting. Radiation-vs-altitude links to aviation dosimetry and cosmic-ray physics. UV-vs-altitude links to public-health UV index forecasts. All three are easy to explain at the presentation and in the outreach report.
 - **Engineering maturity.** The firmware's flight-state machine, FRAM brownout recovery (no lost sequence numbers or flight phase after a reset), hardware watchdog and honest data omission show robust design. Judges also look at reliability and data integrity, not only the science.
 - **Feasible on CanSat hardware.** It needs no extra hardware beyond the sensor suite, stays within the existing radio telemetry format, and its statistics are planned for a short, low-altitude flight. The limits (especially radiation count statistics) are stated up front, as judges expect.
 - **Post-flight deliverable.** Products P1-P7 map directly onto the post-flight results section of the competition's final report and presentation.
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7. Operational requirements for the secondary mission

1. **Power-on $\geq$ 10 minutes before launch.** Lets the MOX sensors (BME688, ENS160, SGP41) complete their warm-up, gives the SGP41 Gas Index algorithm time to start, and allows a long radiation ground baseline. The firmware stays in PRELAUNCH and keeps transmitting during this time.

2. **Record a ≥ 30 min ground radiation baseline** with the same unit on launch day, at the launch site.
3. **Note the conditions:** time, cloud cover, wind and nearby emission sources (vehicles, generators) for the chemistry interpretation.
4. **Mounting:**
 - Keep the gas-sensor inlet clear of the parachute bay and of battery/regulator heat.
 - Mount the VEML6075/OPT3001 behind a UV-transparent window (PTFE diffuser or quartz). Ordinary acrylic/polycarbonate blocks UV-B.
5. **Recovery:** after landing, keep logging for ≥ 2 min in LANDED to get the second ground reference.